

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

to be downloaded to the ATmega16 microcontroller. The file is stored in sub-folder '\default' of your project. If there are error messages, check your 'C' code. Most often these are caused by some typo or syntax errors.

Burn the hex code into the chip using EFY-KnS AVR development board or any other standard AVR programmer/burner.

Construction and testing

An actual-size, single-side PCB layout of the standalone GPS receiver with LCD display is shown in Fig. 4 and its component layout in Fig. 5.

Testing is relatively simple and user-friendly. When you connect 9V power supply to the circuit, LED1 glows, indicating the presence of power supply in the circuit. Wait for some time, say, 5 to 10 minutes, for initialisation of the GPS module.

If everything is fine, you will receive the latitude and longitude right on the LCD screen. Otherwise, switch off the circuit and switch on again, wait for some time and repeat the above steps. You can also press switch S2 momentarily to reset the circuit.

To help you troubleshoot the problems, test points (TP0 through TP5) are indicated in the circuit (Fig. 2). Their details are given in the table.

First, check proper power supply at TP1 with respect to TP0. You should get 9V in your voltmeter. Next, check the voltage at TP2. If you don't get 5V, either 7805 is faulty or there is some connection problem (short or open) in the circuit. Next, check 3.3V at TP3. Otherwise, check the zener diode value and connection. Next, at TP4, check 5V supply for the LCD. Vary the LCD contrast control by turning the preset wiper right and left. Next, check the voltage at TP5. It should be 0V when S2 is pressed, otherwise 5V.

If you still face any problem, please refer to the 'PC-Based GPS Receiver' project published in EFY Oct. 2012 issue. If you are getting proper data on your PC, you will get the same in this project also, provided the gpss.c program is loaded correctly.

Once you have the coordinates on your screen, you can pick up a world map and locate the place yourself. In case you want to see the exact standard time, refer to the time obtained in this application, which entirely depends on satellite navigation. ●

'C' code. If there is no error message, a file called 'gpss.hex' will be generated. This file contains the machine code that is ready



Single Port Ethernet Adaptor

EAD -01


EAD-01, is a low cost Serial RS-232 / RS 422 / RS 485 to Ethernet LAN 10 / 100 Mbps connectivity product using the TCP/IP & UDP protocols and is based on Coldfire high speed 32-bit Microcontroller. EAD has a built in Web Server for configuration through Web Browser and a Factory default switch to load the default configuration settings. The product is available as Module, Board Level and with Plastic & Metal Enclosures.

EAD -01W

Single Port Wireless Ethernet Adaptor



- Secure wireless communication
- Supports range of wireless LAN and security Protocols
- Low power consumption

Two Port Ethernet Adaptor

EAD -02

The EAD 02 connects up to 2 Serial Devices to Ethernet Network using the TCP / IP or UDP protocol. EAD 02 provides Ethernet Connectivity to your products quickly and economically.


EAD -02W

Two Port Wireless Ethernet Adaptor



- Connects two serial devices to a Wireless Network
- End to end security using 256 bit Rijndael AES encryption
- Secure wireless communication
- Extremely effective embedded 802.11 b/g wireless networking

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"An Embedded Systems, Networking & Communication Products Company"

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